AMPLENESS OF AN INVERTIBLE SHEAF PASSES UP AND DOWN ALONG A FINITE SURJECTIVE MORPHISM OF ALGEBRAIC SPACES PROPER OVER A NOETHERIAN RING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- A finite morphism of algebraic spaces is representable by schemes. Morphisms of Spaces, label spaces-morphisms-lemma-integral-local (invoked at proof line 4147). It shows that if X is a scheme then so is Y, which is the scheme-theoretic half of the implication (1) $\Rightarrow$ (2).
- The scheme case of the equivalence for a finite surjective morphism. Cohomology of Schemes, label coherent-lemma-surjective-finite-morphism-ample (invoked at proof line 4149). It supplies ampleness of $f^*\mathcal{L}$ once X is a scheme and $\mathcal{L}$ is ample.
- The cohomological criterion for ampleness, in both directions. Same chapter spaces-cohomology.tex, label lemma-vanshing-gives-ample (invoked at proof lines 4158 and 4191). It converts the property P (vanishing of the cohomology of all twists of every coherent module) into ampleness of $\mathcal{L}$, and over the scheme Y it produces the integer n_0 for the sheaf $i'_*\mathcal{I}'$ twisted by $f^*\mathcal{L}$.
- The criterion reducing a property of coherent sheaves on a Noetherian algebraic space to irreducible supports. Same chapter spaces-cohomology.tex, label lemma-property-higher-rank-cohomological (invoked at proof lines 4159 and 4171). Conditions (1) and (2) are verified directly for P, and condition (3) is met by the sheaf $\mathcal{G} = f'_*\mathcal{O}_{Z'}$ on every reduced irreducible closed subspace.
- The auxiliary finite morphism over an irreducible closed subspace, and the projection formula for an affine morphism. Same chapter spaces-cohomology.tex, labels lemma-finite-morphism-Noetherian (invoked at proof lines 4169 and

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4173) and lemma-affine-morphism-projection-ideal (invoked at proof line 4179). The first provides Z' with the finite morphisms $i' : Z' \to Y$ and $f' : Z' \to Z$ and the support property (3)(a); the second gives $f'_*\mathcal{I}' = \mathcal{I}\mathcal{G}$, which identifies the subsheaf $\mathcal{G}'$ used in (3)(b).

- Coherence of the relevant pushforwards and vanishing of the higher direct images of the finite morphism. Same chapter spaces-cohomology.tex, labels lemma-finite-pushforward-coherent (invoked at proof line 4188), lemma-i-star-equivalence (invoked at proof line 4189) and lemma-finite-higher-direct-image-zero (invoked at proof line 4220), together with the projection formula and the Leray spectral sequence of Cohomology on Sites (proof lines 4216-4219). They let the cohomology of $i_*\mathcal{G}' \otimes \mathcal{L}^{\otimes n}$ on X be rewritten as the cohomology of $i'_*\mathcal{I}' \otimes f^*\mathcal{L}^{\otimes n}$ on the scheme Y, where the vanishing for $n \geq n_0$ is known.

Source reference. Original lemma 0B5V: sources/stacks-chow/tags.txt, source lines 14048–14048. Navigation pointer for \ref{coherent-lemma-surjective-finite-morphism-ample} used inside the counted span (line 4150); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GF8: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3386–3386. Navigation pointer for \ref{lemma-affine-morphism-projection-ideal} used inside the counted span (line 4179); the argument that consumes it is retained in the span.

Source reference. Original lemma 0A4K: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 496–496. Navigation pointer for \ref{lemma-finite-higher-direct-image-zero} used inside the counted span (line 4220); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GF7: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3351–3351. Navigation pointer for \ref{lemma-finite-morphism-Noetherian} used inside the counted span (line 4169); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UH: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2284–2284. Navigation pointer for \ref{lemma-finite-pushforward-coherent} used inside the counted span (line 4188); the argument that consumes it is retained in the span.

Source reference. Original lemma 08AM: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2259–2259. Navigation pointer for \ref{lemma-i-star-equivalence} used inside the counted span (line 4189); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UT: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2652–2652. Navigation pointer for \ref{lemma-property-higher-rank-cohomological} used inside the counted span (line 4159); the argument that consumes it is retained in the span.

Source reference. Original lemma 0GFA: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 4109–4109. Navigation pointer for \ref{lemma-vanshing-gives-ample} used inside the counted span (line 4158); the argument that consumes it is retained in the span.

Source reference. Original section 072X: sources/stacks-chow/tags.txt, source lines 9041–9041. Navigation pointer for \ref{sites-cohomology-section-leray} used inside the counted span (line 4219); the argument that consumes it is retained in the span.

Source reference. Original section 0943: sources/stacks-chow/tags.txt, source lines 11536–11536. Navigation pointer for \ref{sites-cohomology-section-projection-formula} used inside the counted span (line 4218); the argument that consumes it is retained in the span.

Source reference. Original lemma 03ZQ: sources/stacks-chow/tags.txt, source lines 5229–5229. Navigation pointer for \ref{spaces-morphisms-lemma-integral-local} used inside the counted span (line 4148); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a Noetherian ring. Let $f : Y \to X$ be a morphism of algebraic spaces proper over R. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume f is finite and surjective. The following are equivalent*

(1) *X is a scheme and $\mathcal{L}$ is ample, and*
(2) *Y is a scheme and $f^*\mathcal{L}$ is ample.*

Proof. Assume (1). Then Y is a scheme as a finite morphism is representable (by schemes), see Morphisms of Spaces, Lemma 03ZQ. Hence (2) follows from Cohomology of Schemes, Lemma 0B5V.

Assume (2). Let P be the following property on coherent $\mathcal{O}_X$-modules $\mathcal{F}$: there exists an n_0 such that $H^p(X, \mathcal{F} \otimes \mathcal{L}^{\otimes n}) = 0$ for all $n \geq n_0$ and $p > 0$. We will prove that P holds for any coherent $\mathcal{O}_X$-module $\mathcal{F}$, which implies $\mathcal{L}$ is ample by Lemma 0GFA. We are going to apply Lemma 07UT. Thus we have to verify (1), (2) and (3) of that lemma for P. Property (1) follows from the long exact cohomology sequence associated to a short exact sequence of sheaves and the fact that tensoring with an invertible sheaf is an exact functor. Property (2) follows since $H^p(X, -)$ is an additive functor.

To see (3) let $i : Z \to X$ be a reduced closed subspace with $|Z|$ irreducible. Let $i' : Z' \to Y$ and $f' : Z' \to Z$ be as in Lemma 0GF7 and set $\mathcal{G} = f'_*\mathcal{O}_{Z'}$. We claim that $\mathcal{G}$ satisfies properties (3)(a) and (3)(b) of Lemma 07UT which will finish the proof. Property (3)(a) we have seen in Lemma 0GF7. To see (3)(b) let $\mathcal{I}$ be a nonzero quasi-coherent sheaf of ideals on Z. Denote $\mathcal{I}' \subset \mathcal{O}_{Z'}$ the quasi-coherent ideal $(f')^{-1}\mathcal{I}\mathcal{O}_{Z'}$, i.e., the image of $(f')^*\mathcal{I} \to \mathcal{O}_{Z'}$. By Lemma 0GF8 we have $f_*\mathcal{I}' = \mathcal{I}\mathcal{G}$. We claim the common value $\mathcal{G}' = \mathcal{I}\mathcal{G} = f'_*\mathcal{I}'$ satisfies the condition expressed in (3)(b). First, it is clear that the support of $\mathcal{G}/\mathcal{G}'$ is contained in the support of $\mathcal{O}_Z/\mathcal{I}$ which is a proper subspace of $|Z|$ as $\mathcal{I}$ is a nonzero ideal sheaf on the reduced and irreducible algebraic space Z. Recall that f'_*, i_*, and i'_* transform coherent modules into coherent modules, see Lemmas 07UH and 08AM. As Y is a scheme and $\mathcal{L}$ is ample we see from Lemma 0GFA that there exists an n_0 such that

$$H^p(Y, i'_*\mathcal{I}' \otimes_{\mathcal{O}_Y} f^*\mathcal{L}^{\otimes n}) = 0$$

for $n \geq n_0$ and $p > 0$. Now we get

$$\begin{aligned}
H^p(X, i_*\mathcal{G}' \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}) &= H^p(Z, \mathcal{G}' \otimes_{\mathcal{O}_Z} i^*\mathcal{L}^{\otimes n}) \\
&= H^p(Z, f'_*\mathcal{I}' \otimes_{\mathcal{O}_Z} i^*\mathcal{L}^{\otimes n})) \\
&= H^p(Z, f'_*(\mathcal{I}' \otimes_{\mathcal{O}_{Z'}} (f')^*i^*\mathcal{L}^{\otimes n})) \\
&= H^p(Z, f'_*(\mathcal{I}' \otimes_{\mathcal{O}_{Z'}} (i')^*f^*\mathcal{L}^{\otimes n})) \\
&= H^p(Z', \mathcal{I}' \otimes_{\mathcal{O}_{Z'}} (i')^*f^*\mathcal{L}^{\otimes n})) \\
&= H^p(Y, i'_*\mathcal{I}' \otimes_{\mathcal{O}_Y} f^*\mathcal{L}^{\otimes n}) = 0
\end{aligned}$$

Here we have used the projection formula and the Leray spectral sequence (see Cohomology on Sites, Sections 0943 and 072X) and Lemma 0A4K. This verifies property (3)(b) of Lemma 07UT as desired. □